Error for pennylane.numpy.tensor with plugin devices using non-differenti

https://github.com/PennyLaneAI/pennylane/issues/900

ROW 68

Error for pennylane.numpy.tensor with plugin devices using non-differentiable syntax #900

New issue

Closed | #903

antalszava opened on Nov 13, 2020

Issue description

When a `pennylane.numpy.tensor` is passed to a QNode as a keyword argument, it is considered as a non-differentiable parameter. Unlike in the case when it is handled as a differentiable parameter, it is not cast into a `ndarray`.

When indexing into a `pennylane.numpy.tensor`, one can end up having a single element sequence. Certain devices (e.g., `qiskit.aer`, `forest.wavefunction`), are not capable of supporting the application of a gate taking this type of argument and raise an error. Gates can take such a `pennylane.numpy.tensor` as gate parameter, when passed as a non-differentiable parameter.

```
import pennylane as qml
from pennylane.numpy import tensor

dev = qml.device("qiskit.aer", wires=4)

@qml.qnode(dev)
def circuit(phi=None):
    for y in phi:
        for idx, x in enumerate(y):
            qml.RX(x, wires=idx)
    return qml.expval(qml.PauliZ(0))

phi = tensor([[0.04439891, 0.14490549, 3.29725643, 2.51240058]])

circuit(phi=phi)
```

A special case using `RandomLayers` was tackled in PR #893. This is an example that is used in the Quanvolutional Neural Networks demonstration.

- *Expected behavior:*
  The above snippet executes without errors.

- *Actual behavior:*
  Error is raised (Qiskit case):

```
~/anaconda3/lib/python3.7/site-packages/qiskit/providers/aer/backends/backend_utils.py in cpp_execute(co
     49     qobj_dict['config']['library_dir'] = LIBRARY_DIR
     50 
---> 51     return controller(qobj_dict)
     52 
     53 

RuntimeError: Invalid number of dimensions!
```

Further errors may arise for other devices (see #893 (comment) for error messages that were thrown when using `RandomLayers`).

- *Reproduces how often:* (What percentage of the time does it reproduce?)
  For certain plugin devices, each time a `pennylane.numpy.tensor` type sequence is passed as a gate parameter during execution.

- *System information:*

```
Name: PennyLane
Version: 0.13.0.dev0
Summary: PennyLane is a Python quantum machine learning library by Xanadu Inc.
Home-page: https://github.com/XanaduAI/pennylane
Author: None
Author-email: None
```

Assignees

No one assigned

Labels

No labels

Type

No type

Projects

No projects

Milestone

No milestone

Relationships

None yet

Development

Add tensor unwrapping in BaseQNode for keyword arguments
PennyLaneAI/pennylane

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Participants

```
Name: PennyLane
Version: 0.13.0.dev0
Summary: PennyLane is a Python quantum machine learning library by Xanadu Inc.
Home-page: https://github.com/XanaduAI/pennylane
Author: None
Author-email: None
License: Apache License 2.0
Location:
Requires: numpy, scipy, networkx, autograd, toml, appdirs, semantic-version
Required-by: PennyLane-SF, PennyLane-Qchem, PennyLane-Forest, PennyLane-qsharp, pennylane-qulacs, PennyLane-li
Platform info:           Linux-4.19.11-041911-generic-x86_64-with-debian-buster-sid
Python version:          3.7.6
Numpy version:           1.18.5
Scipy version:           1.4.1
Installed devices:
- strawberryfields.fock (PennyLane-SF-0.12.0)
- strawberryfields.gaussian (PennyLane-SF-0.12.0)
- strawberryfields.gbs (PennyLane-SF-0.12.0)
- strawberryfields.remote (PennyLane-SF-0.12.0)
- strawberryfields.tf (PennyLane-SF-0.12.0)
- forest.numpy_wavefunction (PennyLane-Forest-0.13.0.dev0)
- forest.qpu (PennyLane-Forest-0.13.0.dev0)
- forest.qvm (PennyLane-Forest-0.13.0.dev0)
- forest.wavefunction (PennyLane-Forest-0.13.0.dev0)
- microsoft.QuantumSimulator (PennyLane-qsharp-0.9.0.dev0)
- qulacs.simulator (pennylane-qulacs-0.12.0.dev0)
- lightning.qubit (PennyLane-lightning-0.10.0.dev0)
- qiskit.aer (PennyLane-qiskit-0.13.0.dev0)
- qiskit.basicaer (PennyLane-qiskit-0.13.0.dev0)
- qiskit.ibmq (PennyLane-qiskit-0.13.0.dev0)
- default.gaussian (PennyLane-0.13.0.dev0)
- default.mixed (PennyLane-0.13.0.dev0)
- default.qubit (PennyLane-0.13.0.dev0)
- default.qubit.autograd (PennyLane-0.13.0.dev0)
- default.qubit.tf (PennyLane-0.13.0.dev0)
- default.tensor (PennyLane-0.13.0.dev0)
- default.tensor.tf (PennyLane-0.13.0.dev0)
- aqt.noisy_sim (PennyLane-AQT-0.10.0.dev0)
- aqt.sim (PennyLane-AQT-0.10.0.dev0)
- cirq.mixedsimulator (PennyLane-Cirq-0.12.0.dev0)
- cirq.simulator (PennyLane-Cirq-0.12.0.dev0)
```

Source code and tracebacks

Additional information

A solution could be to create a case for `pennylane.numpy.tensor` type sequence parameters in each plugin device implementation (see #893 (comment) containing a suggestion).

Seems that this problem does not arise in tape mode.

josh146 on Nov 13, 2020 Member

Thanks for catching this **@antalszava** **@mariaschuld**.

I think this is a result of the old + new QNode variable marking system clashing.

- Keyword arguments being non-differentiable is only assumed in the old core; PennyLane NumPy tensors are passed as-is to the device if passed as a keyword argument.
- In tape mode, however, positional vs. keyword arg makes no difference. Any argument can be differentiable, if marked as so.

As a result, when tape-mode is default this issue should disappear. However, in the meantime, I suggest we add

```
if isinstance(x, qml.numpy.tensor) and x.ndim == 0:
    x = x.unwrap()
```

either to the (old) autograd interface, the (old) `BaseQNode`, or `QubitDevice`, depending on which one makes more sense; this way we don't have to update all plugins.

```
- aqt.noisy_sim (PennyLane-AQT-0.10.0.dev0)
- aqt.sim (PennyLane-AQT-0.10.0.dev0)
- cirq.mixedsimulator (PennyLane-Cirq-0.12.0.dev0)
- cirq.simulator (PennyLane-Cirq-0.12.0.dev0)
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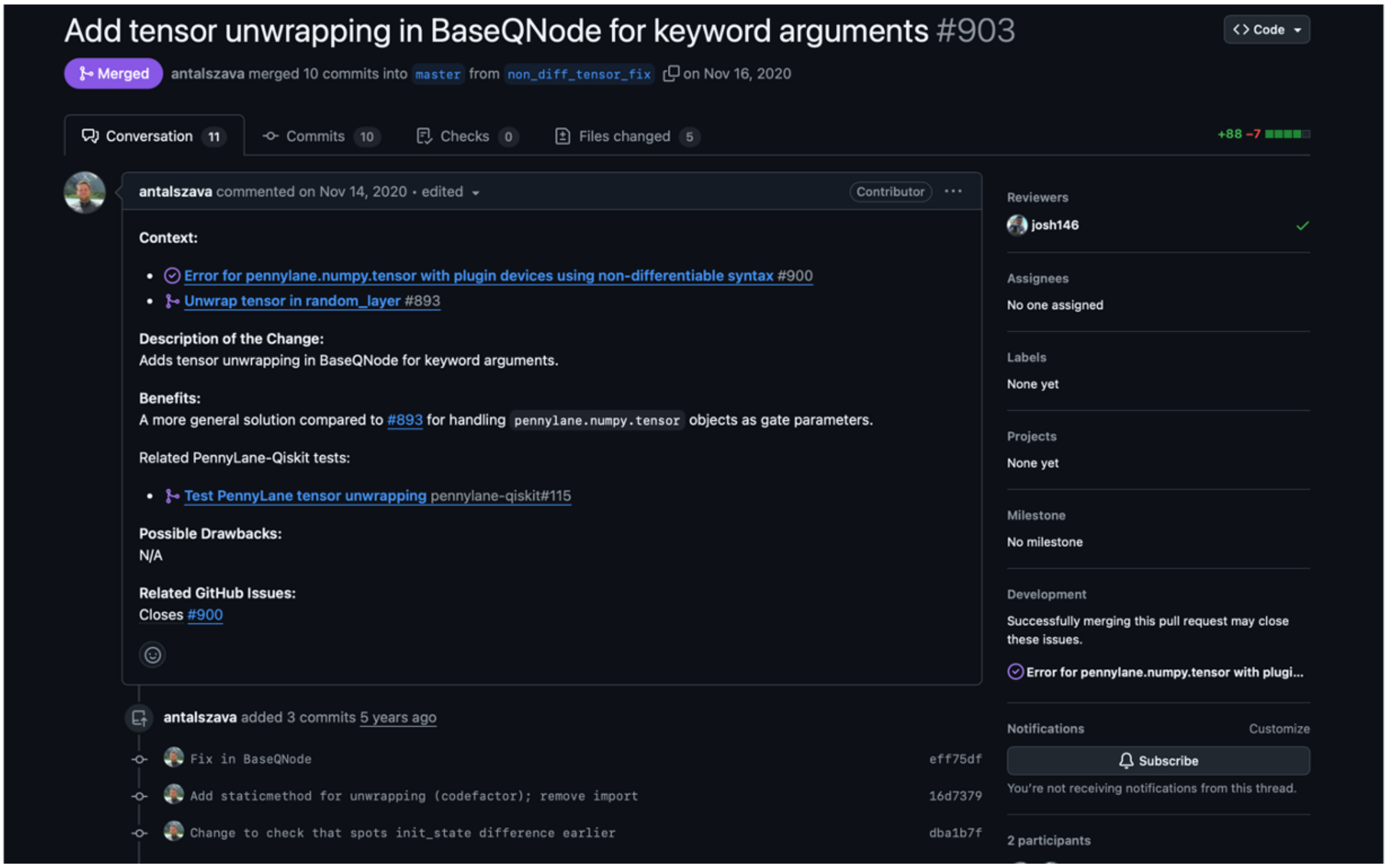
My gut feeling is that it makes more sense to fix this in `BaseQNode`, since it will be deleted anyway when tape mode is default.

antalszava mentioned this on Nov 14, 2020

Add tensor unwrapping in BaseQNode for keyword arguments #903

antalszava closed this as completed in #903 on Nov 16, 2020


Add tensor unwrapping in BaseQNode for keyword arguments #903
Code
Merged
antalszava merged 10 commits into master from non_diff_tensor_fix on Nov 16, 2020
Conversation 11
Commits 10
Checks 0
Files changed 5
+88 −7
antalszava commented on Nov 14, 2020 • edited
Contributor
Context:
Error for pennylane.numpy.tensor with plugin devices using non-differentiable syntax #900
Unwrap tensor in random_layer #893
Description of the Change:
Adds tensor unwrapping in BaseQNode for keyword arguments.
Benefits:
A more general solution compared to #893 for handling pennylane.numpy.tensor objects as gate parameters.
Related PennyLane-Qiskit tests:
Test PennyLane tensor unwrapping pennylane-qiskit#115
Possible Drawbacks:
N/A
Related GitHub Issues:
Closes #900
antalszava added 3 commits 5 years ago
Fix in BaseQNode
eff75df
Add staticmethod for unwrapping (codefactor); remove import
16d7379
Change to check that spots init_state difference earlier
dba1b7f
Reviewers
josh146
Assignees
No one assigned
Labels
None yet
Projects
None yet
Milestone
No milestone
Development
Successfully merging this pull request may close these issues.
Error for pennylane.numpy.tensor with plugi...
Notifications
Customize
Subscribe
You're not receiving notifications from this thread.
2 participants